Unit 9 Pre-Test

Volume of Cylinders, Cones & Spheres

Standard assessed: NC.8.G.9

Name: _____ **Date:** _____ **Period:** _____

SCORE 3.0 Basic Understanding

NC.8.G.9 Find the volume of a cylinder and a cone from the formula.

- *I can plug numbers into $V = \pi r^2 h$ and $V = \frac{1}{3}\pi r^2 h$ to find a volume.*

Match each vocabulary term to its definition, then find each volume.

A radius ✓	**A.** The distance from the center of a circle to its edge.
D diameter ✓	**B.** The amount of space a solid figure holds.
C cylinder ✓	**C.** A solid with two parallel circular bases.
E cone ✓	**D.** The distance across a circle through its center.
F sphere ✓	**E.** A solid with one circular base that narrows to a point.
B volume ✓	**F.** A perfectly round solid, like a ball.

Volume formulas — reference

Cylinder: $\mathbf{V = \pi r^2 h}$ · Cone: $\mathbf{V = \frac{1}{3}\pi r^2 h}$ · Sphere: $\mathbf{V = \frac{4}{3}\pi r^3}$

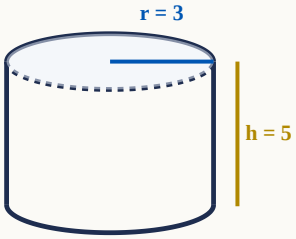
Leave answers **in terms of π** *(for example, 12π) unless told to round.*

1. *Cylinder*

r = 3, h = 5. Find the volume (in terms of π).

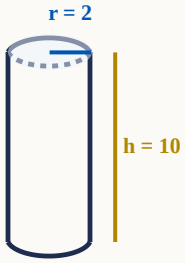
2. *Cylinder*

r = 2, h = 10. Find the volume.



V = _____

answer: 45π

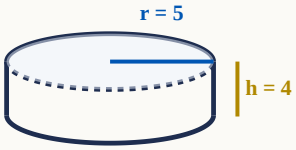


V = _____

answer: 40π

3. *Cylinder*

r = 5, h = 4. Find the volume.

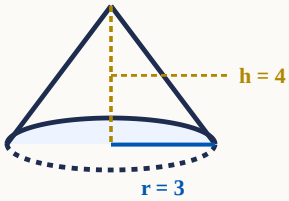


V = _____

answer: 100π

4. *Cone*

r = 3, h = 4. Find the volume.

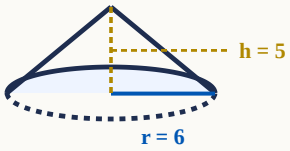


V = _____

answer: 12π

5. *Cone*

r = 6, h = 5. Find the volume.

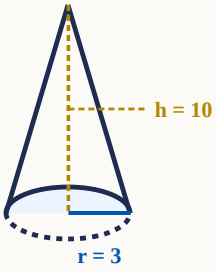


V = _____

answer: 60π

6. *Cone*

r = 3, h = 10. Find the volume.



V = _____

answer: 30π

SCORE **4.0** **Target Content**

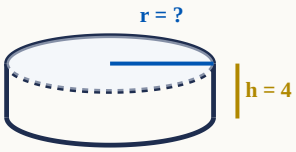
NC.8.G.9 Work backwards from a known volume to find a missing dimension.

• *I can work backward from the volume to find a missing radius or height.*

The volume is given. Find the missing measurement.

7. *Find the radius*

A cylinder has volume 100π and height 4. Find the radius.

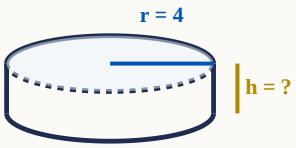


r = _____

answer: 5

8. *Find the height*

A cylinder has volume 48π and radius 4. Find the height.

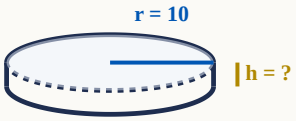


h = _____

answer: 3

9. *Find the height*

A cylinder has volume 200π and radius 10. Find the height.

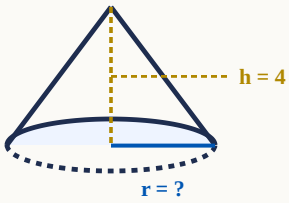


h = _____

answer: 2

10. *Find the radius*

A cone has volume 12π and height 4. Find the radius.

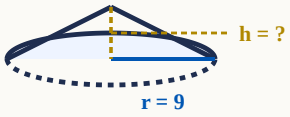


r = _____

answer: 3

11. *Find the height*

A cone has volume 54π and radius 9. Find the height.

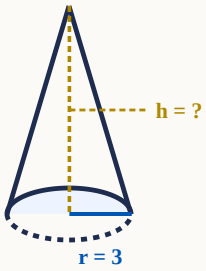


h = _____

12. *Find the height*

A cone has volume 30π and radius 3. Find the height.

answer: 2



h = _____

answer: 10

SCORE 5.0 Advanced Content

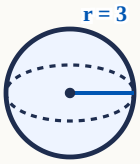
NC.8.G.9 Find the volume of a sphere and solve multi-step problems.

• *I can find the volume of a sphere and compare volumes of different solids.*

Solve each problem. Leave answers in terms of π unless told otherwise.

13. *Sphere*

r = 3. Find the volume.

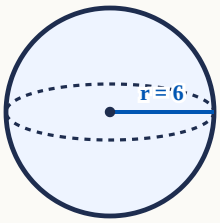


V = _____

answer: 36π

14. *Sphere*

r = 6. Find the volume.

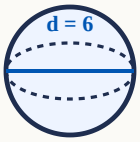


V = _____

answer: 288π

15. *Sphere*

A sphere has diameter 6. Find the volume.

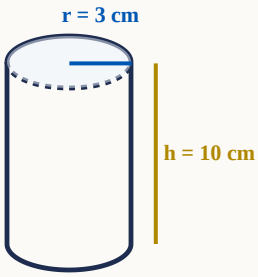


16. *Apply*

A can (cylinder) has r = 3 cm and h = 10 cm. Find the volume in terms of π.

V = _____

answer: 36π

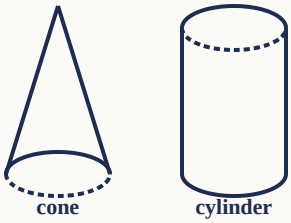


V = _____

answer: 90π cm³

17. *Compare*

A cone and a cylinder have the same radius and height. The cone's volume is what fraction of the cylinder's?



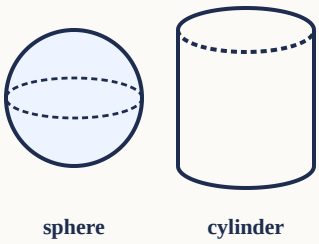
same r, h

Answer: _____

answer: ⅓

18. *Compare*

A sphere has r = 3. A cylinder has r = 3, h = 6. Which has the greater volume?



Answer: _____

answer: the cylinder